

Review retrospective:
*the quartic–sextic moment closed form — one theorem, nothing
to trim*

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Abstract

A soundness review of `Laplace/TwoD/QuarticSexticMoments.lean`: the single closed-form theorem $\langle x^{2j}y^{2k} \rangle_{2D} = ((24/t)^{j/2}\Gamma(\frac{2j+1}{4})/\Gamma(\frac{1}{4}))((720/t)^{k/3}\Gamma(\frac{2k+1}{6})/\Gamma(\frac{1}{6}))$ — the “T1” factorisation that the moment-asymptotic file consumes. Result: **a clean fidelity pass with no edits**. The file has no `open` statements and no dead code; the reliably-safe bucket was empty.

1 Setting

This is the closed-form moment companion to the `QuarticSextic` core: it instantiates the core’s `gibbsExpectation_quarticSextic_factor` at $f = x^{2j}$, $g = y^{2k}$ and substitutes the 1D quartic/sextic even-moment closed forms. Its single theorem is exactly the `gibbsExpectation_quarticSextic_pow_pow_eq` (T1) consumed by the moment-asymptotic specialisation.

2 What was reviewed

One public theorem, `gibbsExpectation_quarticSextic_pow_pow_eq`.

3 Fidelity / soundness findings

Sound; no mismatch. The product closed form is exactly the two 1D even-moment closed forms (`quartic_expected_value_even`: $(24/t)^{j/2}\Gamma(\frac{2j+1}{4})/\Gamma(\frac{1}{4})$; `sextic_expected_value_even`: $(720/t)^{k/3}\Gamma(\frac{2k+1}{6})/\Gamma(\frac{1}{6})$), the $t > 0$ hypothesis is present, and the docstring is appropriately scoped — it states the closed form for $t > 0$, attributes it to factorisation plus the 1D forms, and does not overclaim.

4 Simplifications applied

None — the bucket was empty. The file carries no `open` statements (so no dead open to trim), its single import is load-bearing, and the proof is two genuinely-working `rw` steps. The reviewer’s one suggestion — merging the two consecutive `rw` calls into a single `rw [...] block` — is cosmetic proof-golf, not a dead-code or warning-clearing change, and was declined. No worktree was created.

5 Disagreements and open questions

None.

6 Lessons

- **A two-rw closed-form file is the minimal clean-pass shape.** With no opens, one load-bearing import, and a proof that is exactly “apply the factorisation, substitute the two 1D forms”, there is nothing safe to land — and the cosmetic **rw**-merge is not it. The empty bucket is the result, recorded as such.

7 Follow-ups

- The last `QuarticSextic*` file, `QuarticSexticOddMoments` (the odd-moment vanishing companion).